

■ RULE-BASED PYTHON CHATBOT

30-Minute Student Task • Completed Solution

NAME	REGISTER NO.	SECTION / DEPARTMENT
KODIVALASA ASHIRASHREE	99230040339	S16 • CSE (DS)

■ Objective

Build a simple conversational chatbot that continuously accepts user messages, recognizes greetings and common topics, responds appropriately, and exits when the user types “bye” or “exit”.

■ Chatbot Concept — “StudyBuddy”

StudyBuddy is a friendly student-focused chatbot. It can respond to greetings, study-related questions, motivation requests, and basic Python questions. It also uses flexible keyword matching so phrases such as “hello bot” and “hey hello” are recognized.

■ Python Program

```
print("=====")
print("      ■ STUDYBUDDY CHATBOT")
print("=====")

name = input("Hey! Before we begin, what's your name? ").strip()

print(f"Nice to meet you, {name}! ■")
print("I can help with Python, study tips, motivation, and more.")
print("Type 'bye' or 'exit' whenever you want to leave.\n")

while True:
    message = input(f"{name}: ").lower().strip()

    if message == "bye" or message == "exit":
        print(f"StudyBuddy: Goodbye, {name}! Keep learning and keep growing! ■")
        break

    elif "hello" in message or "hi" in message or "hey" in message:
        print(f"StudyBuddy: Hello {name}! ■ How can I help you today?")

    elif "python" in message or "programming" in message:
        print("StudyBuddy: Python is a beginner-friendly language used for")
        print("web development, automation, data science, AI, and more.")

    elif "study" in message or "exam" in message:
        print("StudyBuddy: Try a focused 25-minute study session,")
        print("take a short break, and then repeat. ■■■")

    elif "motivat" in message or "tired" in message or "stress" in message:
        print("StudyBuddy: Small progress is still progress.")
        print("Take a breath, choose one small task, and start. ■■")

    elif "thank" in message:
        print(f"StudyBuddy: You're welcome, {name}! ■")

    elif "how are you" in message:
        print("StudyBuddy: I'm doing great and ready to chat! ■")

    else:
        print("StudyBuddy: I'm not sure about that yet.")
        print("Try asking about Python, studying, motivation, or say hello.")
```

■ Requirement Checklist

ITEM	IMPLEMENTATION	STATUS
1	Asks the user's name before the main conversation.	✓
2	Uses while True: to keep the conversation running.	✓
3	Uses input() to receive messages.	✓
4	Cleans input using .lower().strip().	✓
5	Recognizes hello, hi, and hey.	✓

ITEM	IMPLEMENTATION	STATUS
6	Answers multiple topics: Python, study/exams, motivation, thanks, and more.	✓
7	Uses if / elif / else for decision making.	✓
8	Uses break for bye or exit.	✓
9	Uses the user's name in responses.	✓
10	Uses else to provide a useful response for unknown messages.	✓
Challenge	Uses the in operator for flexible greeting/topic matching.	✓

■ Suggested Test Run

TEST INPUT	EXPECTED BEHAVIOUR
Name	Ashirashree
Greeting	HELLO BOT
Topic	What is Python?
Study	I have an exam
Motivation	I am tired
Unknown	Tell me a random fact
Exit	bye

■ Target Flow

INPUT → PROCESS → RESPOND → REPEAT → EXIT

Clean • Flexible • Student-friendly • Easy to demonstrate